

ARUN SINGH

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Professional Summary

Mechanical Engineering undergraduate with hands-on experience in manufacturing, industrial systems, and engineering analysis through technical projects. Skilled in process optimization, root-cause analysis, equipment evaluation, and Industry 4.0 technologies. Eager to contribute to production, maintenance, process improvement, and advanced manufacturing while learning and growing in a professional environment.

Education

NMAM Institute of Technology, Nitte	2023 – 2027
B.Tech in Mechanical Engineering	CGPA: 7.66
Class XII, JKBOSE	2023
Bal Shiksha Kender Railway Station, Samba	74.4%
Class X, JKBOSE	2021
Bal Shiksha Kender Railway Station, Samba	84.8%

Skills

Manufacturing & Engineering: Machining, Casting, Assembly, Equipment Maintenance, Process Optimization, Quality Control

Lean / TPS: Lean Manufacturing, 5S, Just-in-Time (JIT)

Tools: MS Excel, AutoCAD, ANSYS, Python

Professional Skills: Leadership, Communication, Teamwork, Adaptability, Time Management

Projects

Precision Micro-Positioning Autofocus Mechanism for Optical Imaging

- Designed a precision micro-positioning mechanism for automated autofocus applications.
- Analyzed positioning accuracy and thermal stability to evaluate thermal drift.
- Developed the CAD design with focus on backlash reduction and precision motion.

Remote Maintenance System for Hydraulics

- Developed an IIoT-based predictive maintenance concept using pressure and vibration monitoring.
- Focused on reducing equipment downtime through early fault detection techniques.
- Analyzed operational data for improving maintenance efficiency and system reliability.

Efficiency Analysis of Industrial Pumping Station

- Evaluated centrifugal pump efficiency and operational reliability using performance analysis models.
- Conducted monitoring and analysis of key operational parameters for process improvement.
- Explored real-time monitoring concepts for improving industrial workflow performance.

Experience

College Intern – HHO Generator Project

Jan 2024 – Apr 2024

- Assisted in research, efficiency analysis, and operational evaluation of an HHO generator system.
- Participated in the study of hydrogen production through water electrolysis processes.
- Contributed to technical documentation and analysis of system performance.

Certifications

- Cognitive Manufacturing – AI and Machine Vision System — L&T EduTech
- Digital Technologies with CPS, IIOT & Cloud Manufacturing — L&T EduTech
- Technology and Processes in Heavy Manufacturing — L&T EduTech
- Advanced Manufacturing: Quality Control Job Simulation — Forage
- Fundamentals of Industry 4.0 and Smart Manufacturing

Leadership

Member, Road Safety Club, NMAMIT

2025 – 2026

- Coordinated awareness initiatives and participated in team-based safety activities.

Member, Institution Innovation Council, NMAMIT

2025 – 2026

- Supported collaborative innovation programs and technical engagement activities.